

A Value-based Maintenance Strategy for Systems under Imperfect Repair and Continuous Degradation

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SUMMARY & CONCLUSIONS

A maintenance optimization model is developed to maximize the net value of imperfect degradation-based maintenance by determining an optimal interval of condition monitoring and the degradation level after imperfect repairs. The proposed decision model intends to maximize the difference between the maintenance value and the expected total cost of maintenance during a life cycle. The value of maintenance depends on the expected life cycle length which positively affected by maintenance activities.

The value-based strategy advocates applying more frequent monitoring. However, the optimal net value might be insensitive to the degradation level after repair. It implies the possibility of reaching the maximum value of maintenance even without the highest level of repairs.

1 INTRODUCTION

Maintenance is a key driver of system availability and profitability. In essence, appropriate management of maintenance requires abundant resources and proper planning. Because of this inevitable use of resources and personnel, maintenance along with other operations activities have been considered as restrictions for more profit, i.e. unwelcomed costs. Operations and maintenance (O&M) Costs contribute to 60–70% of the overall generating cost in nuclear power plants [1] and 14%-30% of the generating cost in offshore wind farms [2]. It almost goes without saying that O&M costs have been under pressure to be minimized. Not surprisingly, existing literature on maintenance modeling and optimization has mostly focused on minimizing maintenance cost [3-7]. In essence, the existing cost-centric models ignore the contribution of maintenance to system value. For instance, improved system reliability is a significant value of maintenance. The idea of pondering maintenance value was first highlighted by Wang [8]. Conceivably, considering maintenance as a value-generating action result in more sophisticated maintenance strategies. However, there is scarce literature on value-based maintenance [9-13].

It should be emphasized that, unlike maintenance cost, it is difficult to formulate maintenance from the value perspective. The complexity of quantifying maintenance value

is compounded by a combination of several factors, the most notable of which are monitoring frequency, monitoring /inspection quality, degradation/maintenance threshold, maintenance degree, multiple interacting components, and optimization criteria.

In this paper, we present a value-based maintenance strategy for a system subject to degradation. The system receives periodic monitoring with constant monitoring interval. The degradation data, such as vibration features, temperature, and oil concentration, can be used to reflect the actual deterioration status. Imperfect maintenance (IM) will be applied for the system under investigation when there is an objective evidence of impending failures. The system condition after maintenance actions returns to somewhere between good-as-new and bad-as-old.

The objective of this research is to maximize the net value of an imperfect degradation-based maintenance strategy by determining the length of the monitoring interval and the degradation level after imperfect preventive repairs (PR). In this respect, a decision model is developed and successfully applied to a group of simulated degradation data.

The remainder of the paper is organized as follows. Section 2 presents the assumptions and the degradation modeling specifications. Section 3 is devoted to deriving the expected life cycle length and the expected total cost of maintenance during a life cycle. The decision model is proposed in Section 4 and followed by a numerical example in Section 5. Finally, concluding remarks are presented in Section 6.

2 DEGRADATION MODELING

In practice, degradation mechanisms can be related to life-time distributions through degradation rate or cumulative degradation models. A variety of degradation models exist in the literature. Ye and Xie [14] provide a comprehensive review of degradation models. The description of our degradation model starts with identifying the basic assumptions.

The following assumptions are used in our modeling:

1. Duration of imperfect repair is negligible.
2. The only cause of system failure is degradation.

3. Degradations before and after repair are independent.
4. Degradation is monotonic and the system experiences only soft failures.
5. Cost of imperfect maintenance depend on the maintenance degree.
6. Degradation threshold are pre-determined. The thresholds are log transformations of original degradation values.
7. Functional failures are detected only by condition monitoring.

We define two thresholds for the degradation model: (i) the threshold between normal state and potential failure (X_p), and (ii) the threshold between potential failure and functional failure (X_f). Accordingly, X_p is the threshold for initiating imperfect preventive repairs (i.e. degradation indicator falls between X_p and X_f). Functional failure happens if the degradation indicator exceeds X_f . Hence, our system needs perfect maintenance, i.e. replacement. Figure 1 shows a typical degradation with one imperfect repair. It should be clear that both value-based and cost-based maintenance highly depend on the threshold values.

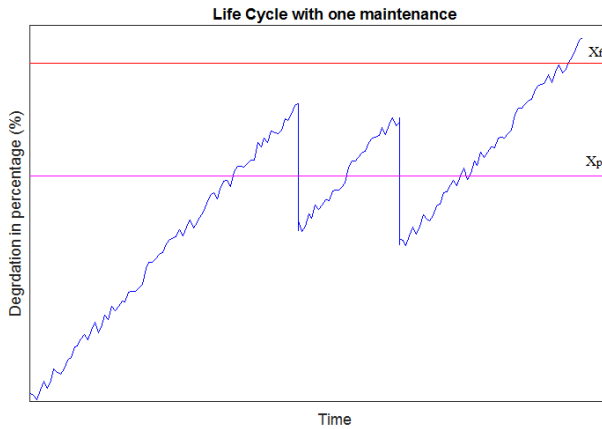


Figure 1. Typical degradation with imperfect repair

In this study, two decision variables are used: (i) the optimal length of monitoring interval, ζ ; (ii) the degradation level after each preventive repair, x_r . It is assumed that after each imperfect repair, e.g. balancing or alignment, the degradation level drops to a fixed value (x_r) between the initial condition and X_p .

The degradation model is based on the model presented in [7] which was initially proposed by Nelson [15]. At time t_i the cumulative degradation, $D(t)$, is expressed as the following exponential form

$$D(t_i) = \Phi + \theta e^{\beta t_i + \varepsilon(t_i)} \quad i = 1, 2, \dots; \quad 0 \leq t_1 \leq t_2 \leq \dots \quad (1)$$

The log form of Equation (1) can be given as

$$L(t_i) = \ln(D(t_i) - \Phi) = \ln\theta + \beta t_i + \varepsilon(t_i) \quad (2)$$

where θ and β are mutually independent and represent characteristics common to all individual systems in the population. Here, $\ln\theta$ has a normal distribution. The error term, $\varepsilon(t_i)$, follows a Markov process and represents the degradation characteristics unique to an individual system. The Markovian property of error terms implies that any two

increments in log degradation are independent from each other and so are the increments in original degradation values. Let $L(t_0) = 0$ and $\theta' = \ln\theta$. Hence, the degradation model in a random coefficient form is given by

$$L(t_i) = \theta' + \beta t_i + \varepsilon(t_i) \quad (3)$$

3 MAINTENANCE MODEL

The value-based decision model has two important elements: (i) the expected length of life cycle, (ii) the expected life cycle maintenance cost. This section explains the process of obtaining these parameters.

3.1 Length of life cycle

It is worth mentioning that the life cycle ends at either a failure or a perfect maintenance, and continues after imperfect maintenance activities. Therefore, the life cycle of a system under investigation consists of three cycles:

1. The first cycle starts from degradation value at time 0 and ends with the first PR.
2. Preventive cycles (PC) start from a just-performed PR and ends with the next PR.
3. Failure cycle starts from a PR and ends with a functional failure.

The life cycle in Figure 1 consists of the first cycle, one PC, and a failure cycle. As mentioned earlier, the system is periodically monitored and PR is performed when the degradation is found to be between X_p and X_f . Consequently, the cumulative degradation descends and subsequently, the degradation process continues in the same life cycle. It is evident that a number of such preventive cycles may happen until a failure is detected. Thus, the set of preventive cycles starts from the first PR and ends at the last PR before the failure cycle. It is clear that the first cycle and the failure cycle are not affected by maintenance activities. In other words, maintenance has no value in these two cycles. Accordingly, from value perspective, the expected length of the life cycle is equal to the expected length of preventive cycles.

Let N_p denotes the number of preventive repairs within a life cycle. Hence, there are $(N_p - 1)$ preventive cycles, and therefore

$$LC(x_r, \zeta) = E[\text{Preventive cycles}] * (E[N_p] - 1) \quad (4)$$

Next, in order to determine the expected life cycle length, we need to derive the probabilities for two cases of degradation process within a life cycle. In case 1, the system will have at least one PR before the failure. Case 2 starts from time 0 and the failure occurs before the first PR. Given the number of condition monitoring periods within a period of time, i , the length of monitoring interval, ζ , and the initial (log) degradation value, x , the probability that a preventive repair occurs before the failure is

$$P_p(i, x, \zeta) = P\{D_1 < x_p, \dots, D_{i-1} < x_p, x_p < D_i < x_f | D_0 = x\} \quad (5)$$

and the probability that a failure occurs before preventive repairs is

$$P_f(i, x, \zeta) = P\{D_1 < x_p, \dots, D_{i-1} < x_p, x_f < D_i | D_0 = x\} \quad (6)$$

As mentioned earlier, D_i denotes the monitored

cumulative degradation. Given $\Delta_j = D_j - D_{j-1}$ (for $J = 1, \dots, i$) as the increment of the degradation values, the probability of having a PR before the failure can be rewritten as

$$P_p(i, x, \zeta) = P\{\Delta_1 < x_p - x, \dots, \sum_{j=1}^{i-1} \Delta_j < x_p - x, x_p - x < \sum_{j=1}^i \Delta_j < x_f - x\} \quad (7)$$

Given $f_{\Delta_i}(\zeta_i)$ as the probability density function (pdf) of the increments, where $i = 1, 2, \dots$, we have

$$P_p(i, x, \zeta) = \left(\int_{-\infty}^{x_p - x} f_{\Delta_1}(\zeta_1) \left(\int_{-\infty}^{x_p - x - \Delta_1} f_{\Delta_2}(\zeta_2) \dots \int_{x_p - x - \sum_{n=1}^{i-1} \Delta_n} f_{\Delta_i}(\zeta_i) d\zeta_{i-1} \dots \right) d\zeta_1 \right) \quad (8)$$

See [7] for more details. By partitioning over all possible number of monitoring intervals, the probability that a PR occurs before the failure can be given as

$$P_p(x, \zeta) = \sum_{n=0}^{\infty} P_p(n, x, \zeta) \quad (9)$$

Similarly, the probability that a failure occurs before preventive repairs is

$$P_f(x, \zeta) = \sum_{n=0}^{\infty} P_f(n, x, \zeta) \quad (10)$$

and obviously,

$$P_f(x, \zeta) + P_p(x, \zeta) = 1 \quad (11)$$

Now, using the probability of PR before failure, the expected length of preventive cycle is

$$E[PC] = \sum_{i=1}^{\infty} (i\zeta) P_p(i, x_r, \zeta) \quad (12)$$

In addition, assuming the number of PR after the initial PR follows a geometric distribution, the expected number of preventive repairs after the initial PR is

$$(Np) \sim \text{Geom}(P_f(x_r, \zeta)) \\ \rightarrow E[Np - 1] = \frac{1}{P_f(x_r, \zeta)} - 1 = \frac{P_p(0; \zeta)}{P_f(x_r, \zeta)} \quad (13)$$

For this distribution, the success is defined as failure occurs before the next PR. Thus, the expected length of a life cycle can be obtained as

$$LC(x_r, \zeta) = \left(\frac{P_p(0; \zeta)}{P_f(x_r, \zeta)} \right) \sum_{i=1}^{\infty} (i\zeta) P_p(i, x_r, \zeta) \quad (14)$$

Then, the overall expected number of condition monitoring intervals during PC can be obtained by

$$E[N] = E[Np + 1] \quad (15)$$

3.2 Life cycle maintenance cost

There are three key components to the life cycle maintenance cost: cost of failure, cost of monitoring or inspection, and cost of PR.

In this study, the cost of PR (C_p) has two elements: a fixed cost (C_s), and a cost associated with degradation reduction: Assume M is a known proportional constant. The cost of a PR at monitoring i can be given as

$$C_p[i] = M(L(t_{i-1}) - x_r)^\alpha + C_s \quad (16)$$

where α is the nonlinearity factor i.e. linear relationship for $\alpha = 1$, and nonlinear for $\alpha > 1$. The expected degradation reduction after a repair, $E[R]$, is needed in order to obtain the overall expected cost of PR, $E[C_p]$. Thus, Equation (16) for

linear relationship is rewritten as

$$E[C_p] = M \cdot E[R] + C_s \quad (17)$$

Based on Equation (3), the expected log degradation level just before a preventive repair at time T is

$$E[L(T)] = \ln\theta + \beta E[T] + E[\varepsilon(T)] = \ln\theta + \beta E(T) \quad (18)$$

Using Equation (12), we have

$$E[L(T)] = \ln\theta + \beta \sum_{i=1}^{\infty} (i\zeta) P_p(i, x_r, \zeta) \quad (19)$$

Thus, the expected degradation reduction is

$$E[R] = \ln\theta + \beta \sum_{i=1}^{\infty} (i\zeta) P_p(i, x_r, \zeta) - x_r \quad (20)$$

This expected degradation reduction is identical for all PCs. Let define C_m as mean cost per monitoring or inspection and C_f as expected cost per failure repair or replacement. Then, the overall expected cost of preventive repairs, during PC, can be obtained as

$$TC(\zeta, x_r) = C_f + E[Np + 1]C_m + E[Np]E[C_p] \quad (21)$$

4 OPTIMIZATION MODEL

In this section, a decision model is proposed to maximize the net value of the maintenance strategy. This decision model can be applied for any engineering system with traceable degradation measures providing that maintenance could induce considerable impact on system degradation. The obvious goal of this research is to highlight the important influence of maintenance on revenue generation. It is absolutely clear that without maintenance the system life is limited to one cycle starting at time zero and ends with a functional failure. Thus, it is reasonable to affirm that the revenue generated during the preventive cycles is the maintenance value. Therefore, the objective function can be defined as the difference between revenue (i.e. maintenance value) and the maintenance cost.

$$\text{Net Value} = \text{Revenue} - \text{Costs}$$

$$Z = V - C$$

It is to be noted that the revenue and cost are associated with the maintenance value and the maintenance costs during PC. In this regard, Equation (21) represents the maintenance costs in the objective function. Moreover, the revenue is defined as

$$V = E[\text{Preventive cycles}] * (E[Np] - 1) * RV$$

or,

$$V = \left(\frac{P_p(0; \zeta)}{P_f(x_r, \zeta)} \right) \sum_{i=1}^{\infty} (i\zeta) P_p(i, x_r, \zeta) * RV \quad (22)$$

where RV is the constant amount of revenue generated in each PC. The variables are the length of the monitoring interval and the degradation level after imperfect preventive repairs. Hence, the decision model can be formulated as

$$\text{Max } Z = V - C$$

$$\text{s.t. } 0 < E[R] < X_p$$

The constraint implies that the degradation level immediately after a repair cannot be greater than the potential failure threshold.

The proposed decision model intends to maximize the

difference between the maintenance value and maintenance cost. In other words, it advocates the functionality of system by providing the optimal length of the monitoring interval and the optimal degradation level after imperfect repair. To this end, one can model the degradation or its log transformed form by a random coefficient function with autocorrelated error terms. Then, continue by estimating the coefficients parameters of the degradation model. The next step is to specify the distribution of degradation error increments and solve for the probabilities of interest. Next, the objective function of the decision model needs to be determined. Note that the constraints of the decision model can be modified with respect to the context of particular applications. Finally, the decision model should be optimized.

5 NUMERICAL EXAMPLE

The decision model developed in Section 4 is applied to a group of simulated degradation data similar to vibration signals. Assume $\mu_\theta = 1$ and $\sigma_\theta^2 = 0.01$. Note that θ is a lognormal variable which follows normal distribution. We use $\beta = 0.125$ and univariate autoregressive-moving-average (ARMA) process for error terms with lag = k (k = 1, 2,...). The error term is centered at zero. In this respect, two settings are examined: lag = 2, and lag = 3. Based on these settings, the log-transformed form in Equation 3 is used to generate a group of 1000 sets of degradation data. For the log degradation data, the threshold of potential failure and the failure threshold are set as 35 and 45, respectively.

The discrete values of the length of monitoring interval and the degradation reduction after preventive repair are as follows

$$\zeta = [25, 50, 75, 100, 125, 150], x_r = [2, 4, 6, 8, 10]$$

The number of monitoring from initial degradation 0 to failure (exceeding X_f) was calculated for each case. The probabilities of interest are obtained based on the frequency count [7].

Table 1. Number of monitoring for lag of 2 and 3

LAG = 2						
Cost		Xr				
		2	4	6	8	10
Zeta	25	84.5	5295.5	5705	5912	6083
	50	63.5	72	199.5	1933.5	2452
	75	50	50	62.5	87.5	212.5
	100	45.5	45.5	45.5	45.5	62
	125	33	53.5	53.5	53.5	53.5
LAG = 3						
Cost		Xr				
		2	4	6	8	10
Zeta	25	75.5	4236.3	4236.3	4236.3	4236.3
	50	52.75	69.25	275.5	4103.5	4103.5
	75	40	52.25	52.25	101.25	554.5
	100	35.75	35.75	35.75	52	68.25
	125	43.75	43.75	64	64	64

Table 1 provides the number of monitoring for lag of 2 and 3 based on different values of x_r and ζ . It is obvious that the life can be extended by performing more monitoring and doing more effective repairs i.e. additional degradation reduction. Interestingly, with the lowest monitoring interval, the lag of two results in more monitoring throughout the life cycle. Generally speaking, increasing the lag leads to decrease in dependence between observations, i.e. the error term in the degradation data.

The values related to the cost model are set as $C_m = 250$, and $C_f = 10000$, $C_s = 2000$, and $M = 2000$. The constant amount of revenue generated in each repair cycle is set as $RV = 200$. Looking at the expected costs based on different values of x_r and ζ in Table 2, it is easy to recognize that the cost-centric strategy would suggest less monitoring and smaller maintenance degree.

Table 2. Expected costs for lag of 2 and 3

LAG = 2						
Number of monitoring		Xr				
		2	4	6	8	10
Zeta	25	17	1175	1266	1312	1350
	50	7	8	23	227	288
	75	4	4	5	7	17
	100	3	3	3	3	4
	125	2	3	3	3	3
LAG = 3						
Number of monitoring		Xr				
		2	4	6	8	10
Zeta	25	18	997	997	997	997
	50	7	9	34	498	498
	75	4	5	5	9	46
	100	3	3	3	4	5
	125	3	3	4	4	4

However, as shown in Table 3, the expected net values are not following this pattern. The results for lag of 2 indicate that for some cases the cost is more than the maintenance value, and therefore the net values can be negative. The highest net value for this setting suggests the highest degradation reduction and more frequent monitoring. This is reasonable since the life is extended with more monitoring and maintenance which results in more revenue. Nevertheless, for lag of 3, the highest net value is for $\zeta = 50$, $x_r = 8$. Interestingly, the net value is the same for $x_r = 8$ and $x_r = 10$. This pattern also happens for $\zeta = 25$ where most of the net values are the same regardless of the degradation reduction. This is an attention-grabbing fact that we can reach the maximum value of maintenance even without the highest level of repairs.

Moreover, it is obvious that the expected net values will increase by increasing the amount of revenue generated in each repair cycle. This can be a motivation to use value-based strategies for critical systems with high revenue such power

generating units.

Table 3. Expected net value for lag of 2 and 3

LAG = 2						
Net Value		Xr				
		2	4	6	8	10
Zeta	25	-4.5	574.5	620	643	662
	50	-3.5	-2	20.5	326.5	418
	75	-5	-5	-2.5	2.5	27.5
	100	-5.5	-5.5	-5.5	-5.5	-2
	125	-8	-3.5	-3.5	-3.5	-3.5
LAG = 3						
Net Value		Xr				
		2	4	6	8	10
Zeta	25	9.5	743.75	743.75	743.75	743.75
	50	7.25	10.75	54.5	866.5	866.5
	75	5	7.75	7.75	18.75	120.5
	100	4.25	4.25	4.25	8	11.75
	125	6.25	6.25	11	11	11

In addition, the change in the failure threshold is also investigated. One can expect that by increasing the failure threshold the number of monitoring will be also increasing. In essence, the number of monitoring is a function of degradation reduction. The results show that the expected number of monitoring will increase for small degradation reduction values and significantly drops with larger degradation reduction values.

The variation in the degradation rate is another subject deserving of attention. Table 4 shows the expected net values as a result of changing β to 0.25 (doubled). This change will increase the degradation rate and the results suggest doing the highest possible repair level with the highest monitoring frequency. This result demonstrates the usefulness of the proposed model.

Table 4. Expected net value for higher degradation rate

Net Value (high degradation rate)		Xr				
		2	4	6	8	10
Zeta	25	-7.75	-6.25	13.25	423.5	431
	50	-8.75	-8.75	-8.75	-7	-5.25
	75	-9.5	-9.5	-9.5	-9.5	-9.5
	100	-8.5	-8.5	-8.5	-8.5	-8.5
	125	-7.5	-7.5	-7.5	-7.5	-7.5

6 CONCLUSIONS

A maintenance optimization model is developed to maximize the net value of imperfect degradation-based maintenance by determining an optimal interval of condition monitoring and the degradation level after imperfect repairs.

The proposed decision model intends to maximize the net value which is the difference between the revenue generated

and the expected total cost of maintenance during a life cycle. The generated revenue in this study depends on the expected life cycle length which positively affected by maintenance activities. The proposed model would be beneficial for engineering systems using costly imperfect maintenance such as bridges and wind turbines.

With no doubt, longer monitoring intervals increase the risk of failure i.e. risk of shorter useful life. For this, the value-based strategy advocates taking more frequent monitoring. However, the optimal net value might be insensitive to the degradation level after repair. It implies the possibility of reaching the maximum value of maintenance even without the highest level of repairs.

In effect, the results indicate the importance of the decision variables for value-based maintenance strategies. Moreover, the results indicate direct relationship between the repair level and the expected cost. Furthermore, from the results it is apparent that the thresholds can play a decisive role in the success of such decision models.

To further improve the model, there are a number of important factors to consider including: rate of degradation, accurate calculation of the cost associated with the degradation reduction, and revenue calculation especially in the case of partial failure. In addition, relaxing the limiting assumptions might be a great challenge for future works. In particular, we need to consider shocks as possible causes of failure and the duration of maintenance based on the required repair level.

Finally, applying an appropriate optimization method is of utmost importance. The response surface method (RSM) seems to be a good candidate to solve the optimization model. Although RSM is effective in searching heuristic optimal solutions for response with complex forms, the optimal value derived by RSM is not necessarily a global optimal value.

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